

Dvpr

Design Verification Plan and Report (Virtual)

Virtual Battery Factory · case t7_r1 · 2026-08-25

Design Verification Plan and Report (Virtual Test) — Case t7_r1

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#	Item	Condition	Result	Determination	Source
1	1C discharge capacity	1C (6.0313 A nominal) discharge, 298.15 K, DFN	6.032209 Ah	PASS (matches nominal 6.0313 Ah within 0.02 %)	cell/r5_v11_1c_dfn.json:capacity_ah
2	Energy density	contract formula: $\int V \cdot I \, dt \div \Sigma \text{layer thickness} \times (1 - \epsilon) \times \rho \times \text{area}$ (electrolyte excluded)	483.247 Wh/kg	PASS vs threshold ≥ 327.18 Wh/kg	cell/r5_v11_energy.json:energy_density_wh_kg
3	4C fast-charge plating	4C charge at 45 °C ambient, lumped thermal + plating model, DFN	anode surface potential min +0.047604 V $\rightarrow$ plated = false	PASS (no potential < 0 V)	cell/r5_v11_4c45.json:anode_potential_v
4	4C fast-charge temperature	same run	T_max = 330.130 K (56.98 °C)	informational — no T_max threshold in task (entry 0 "not_adjudicated")	cell/r5_v11_4c45.json:T_max_K
5	SEI thickness after 100 cycles @ 45 °C	aging_1C_100cyc_45C protocol, isothermal + SEI (ec reaction limited)	465.473 nm	PASS vs threshold ≤ 550 nm	cell/r5_v11_aging45.json:sei_thickness_nm_end
6	Nail penetration $\rightarrow$ thermal runaway (virtual)	run-tr ODE, q_nail = 10 W, hA = 0.2655 W/K (= h 50 $\times$ 0.00531 m²), mass 45.105 g	triggered = false; T_max 336.320 K; dT/dt max 0.244883 K/s	PASS (no TR)	cell/r5_v11_nail.json
6a	Nail cooling-margin sensitivity	h=40 (hA 0.2124): not triggered, T_max 348.69 K; h=30 (hA 0.1593): TRIGGERED at 666.0 s; h=25 (hA 0.1328): TRIGGERED at 518.4 s	design cooling floor ≈ 40 W/(m²·K)	margin evidence for h = 50 design point	cell/r4_v8_nail.json; cell/r5_v8_nail_h30.json; cell/r5_v8_nail_h25.json
7	Voltage window	parameter set upper/lower cut-off	2.5 – 4.2 V	PASS (as configured)	Chen2020 parameter set
8	Baseline reference (R1)	Chen2020 unmodified, same protocols	ED 400.75 Wh/kg PASS; SEI 476.1 nm PASS; 4C plated = true (min -0.1918 V) FAIL; nail triggered at 322.2 s FAIL	baseline failures localized to plating + nail — both fixed by design	log.jsonl evaluate R1 evidence

N/A Items (beyond pure-simulation boundary — require physical experiment)

Item	Status
Physical nail penetration / crush / drop	N/A (virtual TR ODE only; mechanical abuse requires hardware test)
Overcharge $\rightarrow$ thermal runaway	N/A (overcharge protocol not run — overcharge not part of task criteria)
Cycle life to failure (capacity-based EOL)	N/A beyond 100 simulated cycles
Rate-pulse internal resistance (HPPC)	N/A (DCR reported from discharge calc instead: 2.882 mΩ)

Conclusion

PASS — 4/4 adjudicable criteria met at DFN precision (energy density, plating-free 4C, SEI ≤ 550 nm, no nail TR). Note on aging trajectory: per-cycle capacities in the aging protocol are distorted by the thick-SEI-film model artifact at μm -scale films (baseline and all variants affected); the adjudicable metric per entry 0 is sei_thickness_nm_end (465.47 nm $\leq$ 550 nm), and the trajectory is labeled honestly in evaluate R3–R5 notes rather than interpreted as capacity behavior.